Project Progress Presentation on

Real-Time Dynamic Path Planning via Learned First-Order Perturbations of Laplacian Potential Fields

Boxiang (William) Fu

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Motivation

Two equally valid descriptions of the universe...

Newtonian Mechanics

- Local approach
- Mediated by Force particles
- Solves for cause-and-effect one step at a time

Lagrangian Mechanics

- Global approach
- Mediated by Force fields
- Solves for entire trajectory using the Principle of Least Action

Robotics Application

- Dijkstra's and A* algorithms
- Graph traversal algorithms

Robotics Application

- Navigation functions
- Laplacian potential fields

Problem Statement

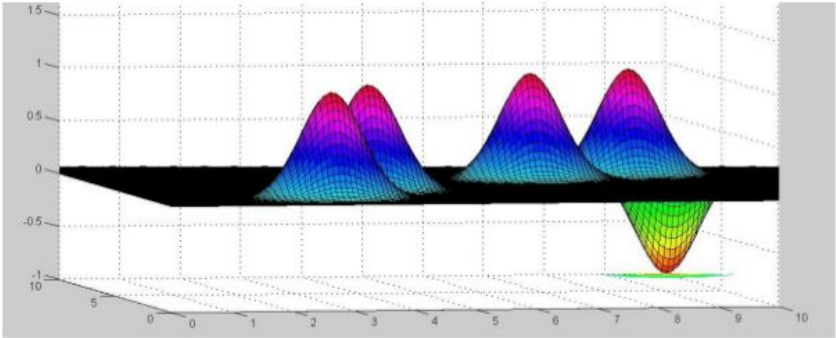
- Given an initial static map with unknown dynamic obstacles, solve the entire (global) trajectory in real-time to the goal location
- Use case: Shopping malls with known floor layout and dynamic pedestrians. Robot must reach goal location while avoiding pedestrians

Related Work

- Path planning via navigation functions (Rimon & Koditschek, 1988)
- Laplace's equation as navigation function (Connolly et al., 1990)
- Heat transfer equation as navigation function (Wang and Chirikjian, 2000)
- Path planning via Hamiltonians (Casagrande et al., 2017)
- Artificial potential fields (Wu et al., 2023)
- Dynamic path planning via perturbed Laplace's equation (Ours)

Methodology (Static Map)

- Model static obstacles and map boundaries as Neumann boundary conditions (i.e. normal derivative is 0)
- Model source location as positively charged source particle
- Model goal location as negatively charged sink particle
- Solve for Laplace's equation in free-space ($\nabla^2\varphi(x) = 0$)
- Robot will follow the path of least resistance ($dx/dt = -\nabla\varphi(x)$)

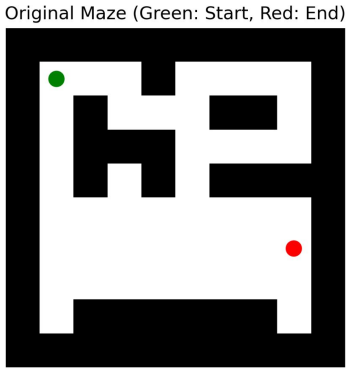


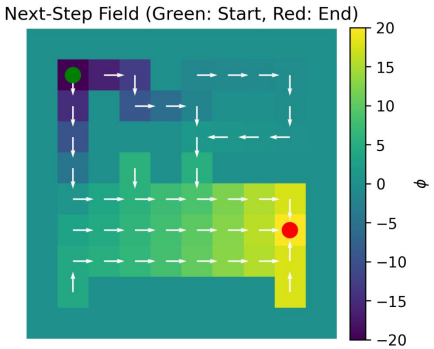
Methodology (Dynamic Map)

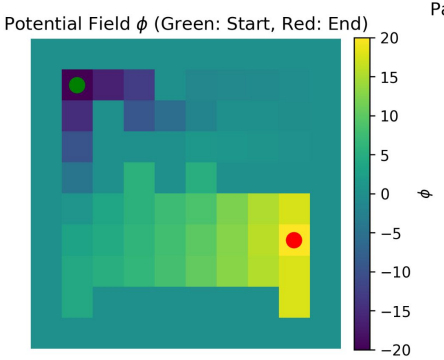
- Look for disturbances in the local neighborhood of the robot
- Model additional dynamic obstacles as repulsive charged particles
- Impose Dirichlet conditions equal to the original potential field at the boundary of local neighborhood
- Use neural network to estimate the first-order perturbation in the potential field due to the new particles in the local neighborhood
- Superimpose the first-order perturbation with the original potential field
- Robot will follow the (new) path of least resistance ($dx/dt = -\nabla\varphi(x)$)
- Repeat for next timestep

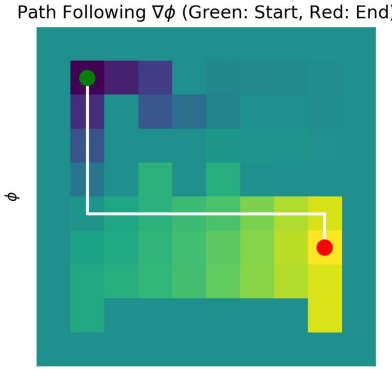
Progress – Static Potential Map

- Solves static Laplace's equation using sparse linear system Aφ = b
- Encode neighbor relationships and permissible flux flow cells in A
- Encode boundary conditions and charge sources/sinks in b
- Solve linear system using SciPy sparse linear solver
- Path following algorithm calculates local gradient at location of robot
- Take step in direction of gradient

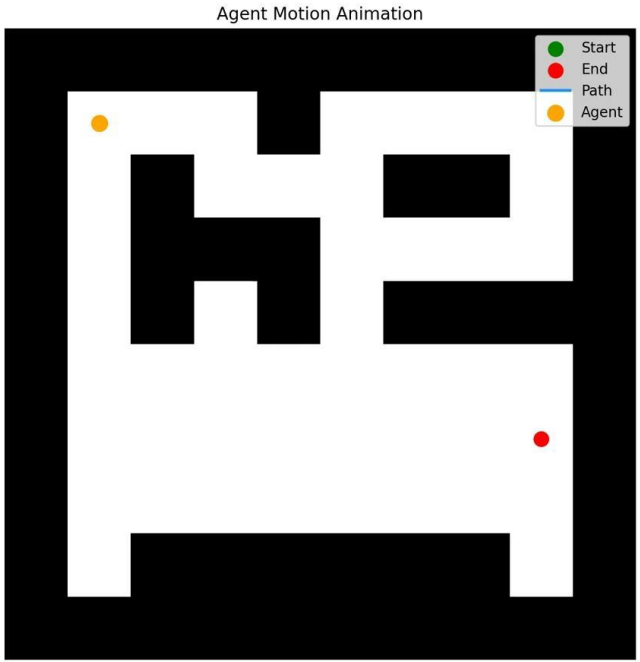




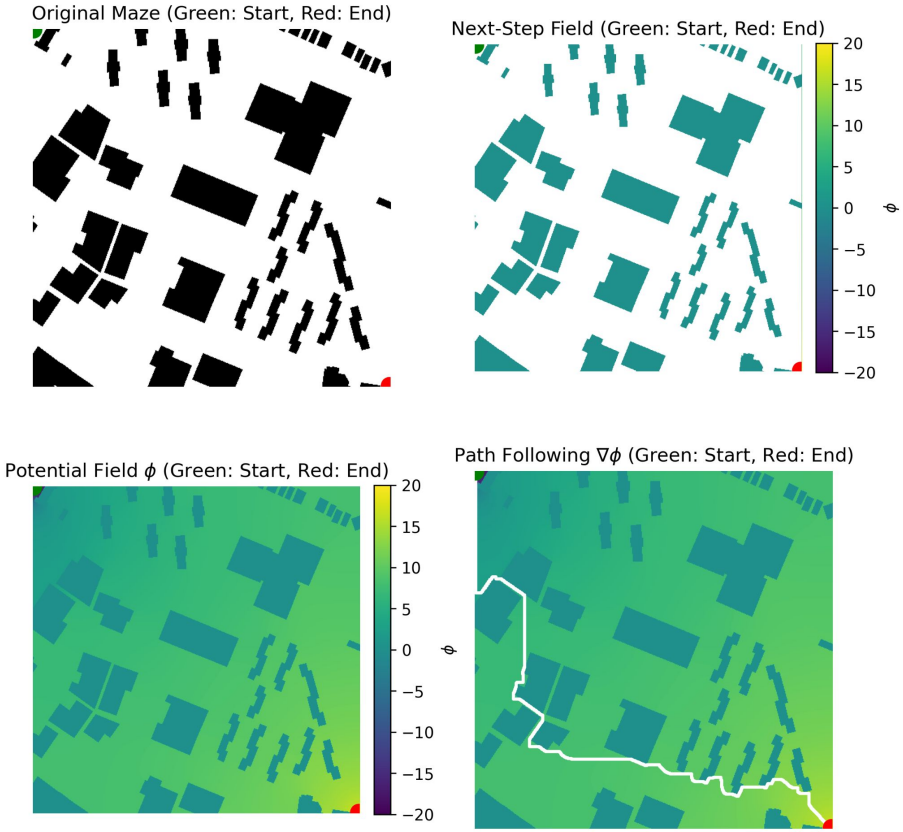




Progress – Animation

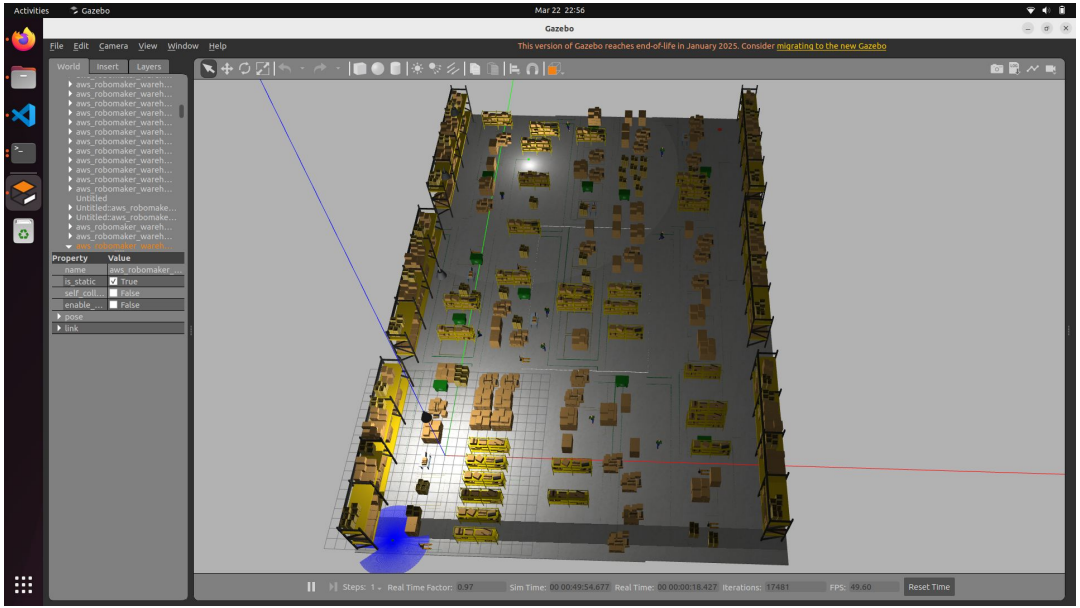


Progress – New York 512x512 Grid Map



Progress – Simulation Setup

- Gazebo simulator of warehouse map with moving humans
- Robot modelled as TurtleBot Burger with 360 degree FOV within a certain radius
- Warehouse modelled by extending Logistics_Warehouse from AWS Robomaker



Insights Until Now

- Iterative Laplace solvers scales quadratically with map dimensions, takes too long to solve for large maps
- Numerical errors from iterative solvers can cause dead ends and incorrect path following
- **Solution: Relaxation Method → Sparse Linear Solver Method**
- Dirichlet boundary conditions can sometimes cause agent to get “sucked” into a wall or dead end
- **Solution: Dirichlet B.C. → Neumann B.C.**

Future – Learned Dynamic Obstacles

- Black-box neural network function
- Input:
  - Current robot location
  - Neighborhood radius
  - Change in map of local neighborhood
  - Value of potential field at boundary of neighborhood
- Output:
  - First-order perturbed potential field in local neighborhood

Future – Simulation Deployment

- Port existing algorithms from 2D grid map to simulator
- Discretize warehouse map and solve for Laplace's equation in the simulator
- Deploy static pathfinding algorithm in simulation
- Robot sometimes gets stuck on "air walls", need to find bug and fix
- Implement local neighborhood object detection for the TurtleBot
- Port learned dynamic obstacles function from 2D grid map to simulator
- Deploy dynamic pathfinding algorithm in simulation

Expectations of Final Result

- Path planning algorithm outputs a static potential map of the known environment
- Perturbation algorithm outputs the first-order perturbation of the potential field in the local neighborhood of the robot
- Path following algorithm is expected to follow the path of least resistance ($dx/dt = -\nabla\varphi(x)$) and should not hit dynamic obstacles
- The algorithm should be deployed in the Logistics_Warehouse environment using the Gazebo simulator

Final Timeline

- (Week 10) A learned neural network that estimates the first-order perturbation in the potential field due to nearby obstacles
- (Week 11) A dynamic path planning algorithm using the perturbed Laplacian potential field in a dynamic 2D environment
- (Week 13) Deploy the algorithm in a realistic simulator with dynamic obstacles
- (Week 14) Final write-up and presentations

Thank You!